Constraint-Preserving QAOA for Flood Shelter Assignment: A Statevector Study on Digital-Twin Instances

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***Abstract*—Assigning evacuation groups to shelters under road closures and limited capacity is a small constrained combinatorial problem, and it is often cited as a target for quantum optimization. We test this claim on instances taken from a flood digital twin rather than on random graphs. Each instance assigns four groups to three shelters after a road closure and a shelter surge, giving a 12-qubit quadratic unconstrained binary optimization (QUBO) problem. We compare the Quantum Approximate Optimization Algorithm (QAOA) with the standard transverse-field mixer against a constraint-preserving ring-XY mixer, each trained on the mean energy and on the conditional value-at-risk (CVaR), at depths 1 to 4, using exact noiseless statevector simulation. On 30 feasible instances, the XY mixer placed 0.215 of its probability on the optimal plan at depth 4, compared with 0.040 for the standard mixer and 0.0169 for uniform sampling of one-hot plans, and it kept most of its probability on feasible plans at every depth. CVaR training raised the chance of finding the optimum within a fixed number of shots for both mixers. In a gate-level simulation with a 1% two-qubit Pauli error rate, the XY mixer still reached $P_{\mathrm{opt}} = 0.105$, against 0.033 for the standard mixer. Classical methods remain far ahead at this size: exhaustive search solved every instance in 0.90 ms. A scaling run up to 18 qubits shows how quickly simulation cost grows. We claim no quantum advantage. The study shows which design choices matter if such problems are moved to quantum hardware.**

***Index Terms*—QAOA, quantum alternating operator ansatz, XY mixer, QUBO, CVaR, evacuation planning, shelter assignment, digital twin**

I. Introduction

When a flood closes roads and fills shelters, an emergency officer must decide again which shelter each evacuation group should go to. The decision couples travel time, flood exposure along the route and the seats left at each shelter. Written as a binary program, it becomes a constrained assignment problem of the kind that is routinely encoded as a QUBO or Ising model [1], [2] and offered as an application for near-term quantum optimization [3], [4].

Two questions are usually left open in such proposals. The first is whether the quantum circuit respects the constraints or merely penalises their violation. In the original QAOA [5], constraints enter the cost function as penalties and the transverse-field mixer explores all 2^n bitstrings, most of which are not valid plans. The quantum alternating operator ansatz [6] instead uses mixers that keep the state inside the feasible subspace of one-hot constraints, such as XY mixers [7], [8]. The second question is how the result compares with ordinary classical methods on instances of realistic shape. Benchmarks on random graphs do not answer it for evacuation.

We address both questions on instances generated by the digital twin of a flood decision-support prototype. Our contributions are:

1) A QUBO formulation of shelter assignment built from twin state, with a pairwise capacity encoding, a penalty bound that makes every violation more expensive than any feasible plan, and a measured exactness rate for the encoding.
2) A controlled comparison of the transverse-field and ring-XY mixers, under mean-energy and CVaR training [9], at depths 1–4 on 30 instances, using exact statevector simulation.
3) A scaling study from 6 to 18 qubits and a comparison with exhaustive search, simulated annealing [10] and a greedy rule.

All results are from classical simulation of noiseless circuits. We make no claim of quantum speed-up, and the classical baselines are reported because they are faster.

II. Related Work

Formulations. Lucas gave Ising forms for many NP-hard problems [1], and Glover *et al.* describe how assignment and capacity constraints become quadratic penalties [2]. Shelter location and allocation for flood evacuation has long been studied with classical integer programming [11]. Quantum annealing has been applied to disaster resource allocation [3].

QAOA and mixers. QAOA alternates a cost unitary and a mixing unitary for p layers [5]. Hadfield *et al.* generalised the mixer so that it preserves hard constraints [6]. Wang *et al.* analysed ring and complete XY mixers for one-hot constraints [7], and Fuchs *et al.* gave a general construction of constraint-preserving mixers [8]. Preparing the initial superposition of one-hot states requires W-state circuits [12].

Training and limits. Zhou *et al.* proposed interpolating optimal angles from depth p to $p{+}1$ (INTERP) [13], and Barkoutsos *et al.* showed that optimising the CVaR of the energy, rather than its mean, increases the probability of good samples [9]. Warm starts from classical solutions are another option [14]. Most published comparisons of mixers and objectives use MaxCut or random constrained instances. Shelter assignment differs in two ways that matter for QAOA: its costs come from travel times that are close to each other,

so the optimum is separated from many near-optimal plans by small energy gaps, and its capacity constraints couple groups that would otherwise be independent. We therefore evaluate on instances produced by the same digital twin that drives the decision-support prototype, after the same disruption the twin uses in its routing benchmark. On current processors, QAOA performance falls with problem size and circuit depth [15], and reviews of noisy intermediate-scale quantum (NISQ) algorithms stress that practical advantage in optimization has not been shown [4], [16], [17].

Research gap. The mixer and objective studies above use MaxCut-style or random instances, and the disaster-response work [3] solves a penalty QUBO on an annealer without comparing formulations. To our knowledge, constraint-preserving QAOA has not been evaluated on capacity-constrained evacuation instances against an exact optimum, classical baselines and a gate-level noise model together. This paper provides that evaluation.

III. Problem Formulation

A. Instances from the Digital Twin

The twin models a corridor with three population zones, nine junctions and three shelters joined by 22 roads. Every road carries a travel time, a flood-depth index and a congestion level drawn from a fixed seed. Shelters have capacities of 120, 90 and 70 people with an initial occupancy of 15–45%, and groups have 8–22 people. Each instance applies the disruption used in the twin's routing benchmark: road R5 closes and Shelter A receives arrivals equal to half its capacity. For group g and shelter s we take the best open path under the twin's flood- and congestion-weighted edge cost, and define

$$c_{gs} = T_{gs} + 5H_{gs}, \tag{1}$$

where T_{gs} is travel time in minutes and H_{gs} is the mean flood index along the path. The pair (g, s) is *forbidden* if no open path exists, the shelter has failed, or the group alone exceeds the remaining seats R_s. With G groups, binary variables x_{gs} give $n = 3G$ qubits. A plan is feasible if every group has exactly one shelter, no forbidden pair is used and $\sum_g n_g x_{gs} \le R_s$ for each shelter, where n_g is the group size.

B. QUBO with Pairwise Capacity

The capacity constraint is linear in x but has an inequality, which normally needs slack qubits [2]. We use a pairwise encoding instead. Let $\mathcal{O} = \{(g, h, s) : n_g + n_h > R_s\}$ be the pairs of groups that cannot share shelter s. With shifted costs $c'_{gs} = c_{gs} - \min_{s'} c_{gs'}$, the energy is

$$E(\mathbf{x}) = \sum_{g,s} c'_{gs} x_{gs} + P\Big[\sum_{g}\Big(1 - \sum_{s} x_{gs}\Big)^2 + \sum_{(g,s)\in\mathcal{F}} x_{gs} + \sum_{(g,h,s)\in\mathcal{O}} x_{gs}x_{hs}\Big], \tag{2}$$

where $\mathcal{F}$ is the set of forbidden pairs.

Lemma 1: With $P = 1 + \sum_g \max_s c'_{gs}$, every bitstring that breaks a one-hot, forbidden-pair or pairwise-capacity term has higher energy than every bitstring that breaks none of them.

Proof. Since $c'_{gs} \ge 0$, a bitstring with no violated term has energy at most $P - 1$. A violated term adds at least P, and the cost part is non-negative. ∎

On one-hot states, the shift subtracts the constant $\sum_g \min_s c_{gs}$, so the optimal plan is unchanged, but the energy scale now reflects only the decision being made. Without the shift, the penalty dominated the energy and QAOA learned feasibility but not cost in our preliminary runs. The pairwise encoding is exact when no three groups overload a shelter that each pair could share. When that fails, the QUBO minimiser can be infeasible, so every plan is re-checked exactly before use. We report how often the minimiser of (2) is the true optimum.

IV. QAOA Variants

For depth p and angles $(\boldsymbol{\gamma}, \boldsymbol{\beta})$, QAOA prepares

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{k=1}^{p} e^{-i\beta_k B}\, e^{-i\gamma_k \hat{E}/\Delta}\, |\psi_0\rangle, \tag{3}$$

where $\hat{E}$ is diagonal with entries $E(\mathbf{x})$ and $\Delta = P - 1$ normalises feasible energies to $[0, 1]$. Substituting $x_q = (1 - Z_q)/2$ turns (2) into an Ising Hamiltonian

$$\hat{E} = \sum_q h_q Z_q + \sum_{q<r} J_{qr} Z_q Z_r + \text{const}, \tag{4}$$

so the cost unitary needs one single-qubit R_Z rotation per qubit and one two-qubit ZZ rotation per non-zero J_{qr}. Each quadratic term of (2), whether from the one-hot penalty or from an overloaded pair, therefore adds a two-qubit gate to every layer.

Transverse-field (X) mixer. $B = \sum_q X_q$ and $|\psi_0\rangle = |+\rangle^{\otimes n}$. All 2^n bitstrings are reachable, and constraints act only through the penalties in (2).

Ring-XY mixer. For each group g with qubits a_g, b_g, c_g,

$$B_{\mathrm{XY}} = \sum_g \sum_{(u,v)\in\{(a_g,b_g),(b_g,c_g),(c_g,a_g)\}} \tfrac{1}{2}\big(X_u X_v + Y_u Y_v\big). \tag{5}$$

Each term swaps amplitude between $|01\rangle$ and $|10\rangle$ and conserves the number of ones, so a one-hot group stays one-hot [7]. The start state is a product of W states, $|W_3\rangle^{\otimes G}$ with $|W_3\rangle = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$, which is the uniform superposition of the 3^G one-hot plans. The one-hot penalty is then constant and is dropped, which also removes $3G$ two-qubit cost terms per layer. The three ring terms do not commute and are applied in a fixed order, as in a first-order Trotter step.

Training. We minimise either the mean energy $\langle \hat{E} \rangle$ or the conditional value-at-risk with $\alpha = 0.1$ [9]. Sorting bitstrings by energy, $E_{(1)} \le E_{(2)} \le \ldots$, with measurement probabilities $\pi_{(k)}$ and cumulative mass $\Pi_k = \sum_{j\le k} \pi_{(j)}$,

$$\mathrm{CVaR}_\alpha = \frac{1}{\alpha} \sum_k E_{(k)}\, \min\big(\pi_{(k)},\ \max(0,\ \alpha - \Pi_{k-1})\big), \tag{6}$$

Algorithm 1 Depth-progressive QAOA training

build $\hat{E}/\Delta$ from (2); choose mixer B and $|\psi_0\rangle$
$\boldsymbol{\theta}^{(0)} \leftarrow \emptyset$
for $p = 1$ **to** 4 **do**
  starts $\leftarrow$ two draws $\gamma_k \sim U(0, \pi)$, $\beta_k \sim U(0, \pi/2)$
  if $p > 1$ **then**
    add INTERP($\boldsymbol{\theta}^{(p-1)}$) to starts
  end if
  $\boldsymbol{\theta}^{(p)} \leftarrow$ best COBYLA result over starts on $\langle \hat{E} \rangle$ or (6)
  record P_{opt}, P_{feas} and evaluations for $\boldsymbol{\theta}^{(p)}$
end for

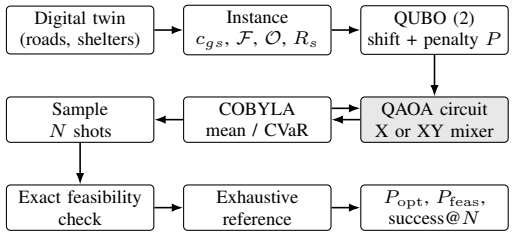


Fig. 1. Experimental pipeline. The shaded block is simulated exactly as a statevector. Every sampled plan is checked against the exact constraints, and the exhaustive reference gives the optimum.

the mean energy of the lowest α probability mass. Minimising it rewards the circuit for placing some probability on low-energy plans even when the rest of the distribution is poor. Angles are optimised by COBYLA [18] with at most 250 evaluations per start. At $p = 1$ we use two random starts. At $p > 1$ we add an INTERP start built from the $p - 1$ optimum [13] to two random starts and keep the best (Algorithm 1). Fig. 1 summarises the pipeline.

A. Simulator

The simulator stores the 2^n complex amplitudes in double precision and applies each layer directly. The cost unitary is diagonal, so it is one elementwise multiplication by $e^{-i\gamma_k E(\mathbf{x})/\Delta}$ with the energies of all bitstrings computed once per instance. The transverse-field mixer applies $R_X(2\beta_k)$ to one qubit at a time by reshaping the amplitude vector so that the target qubit becomes a length-2 axis. Each XY term mixes only the amplitude pairs whose two target bits are 01 and 10; we precompute those index pairs once and update them with a 2×2 rotation. One objective evaluation at 12 qubits takes under ten milliseconds on a laptop, so the cost of training is dominated by the number of objective evaluations, which we report. All random choices (instances, restarts, sampling and noise) use fixed seeds.

V. Experimental Setup

We generated 30 instances from the twin with seeds $42+17i$, $i = 0, \ldots, 29$, and used the first four groups ($G = 4$, $n = 12$). All 30 instances had at least one feasible plan, on average 45.0 of the 81 one-hot plans. In 12 instances, sending every group to its cheapest shelter overloaded a shelter, so capacity changed the optimal plan. All circuits were simulated as exact 2^n-amplitude statevectors in double precision, without noise or shot noise during training.

We report three metrics: P_{opt}, the probability of measuring an optimal plan; P_{feas}, the probability of measuring a feasible plan; and success@N, the probability that at least one of $N = 32$ shots is optimal, $1 - (1 - P_{\text{opt}})^N$. We also count objective evaluations, since each one corresponds to a batch of circuit executions on hardware. The baselines are uniform sampling of all 2^{12} bitstrings ($P_{\text{opt}} = 0.00033$ on average) and of the 81 one-hot plans (0.0169); exhaustive search; single-bit-flip simulated annealing on (2) with one restart of 60 sweeps; and a greedy rule that places the largest group first in its cheapest shelter with seats left. The scaling study used the first ten instances with $G = 2, \ldots, 6$ ($n = 6, \ldots, 18$), the mean objective and $p = 2$.

VI. Results

A. Mixer and Objective

Table I and Fig. 2 give the main results. The XY mixer was better on both metrics. At $p = 4$ with the mean objective it reached $P_{\text{opt}} = 0.215$, which is 12.7 times uniform one-hot sampling and 5.3 times the X mixer at the same depth. Because its state never leaves the one-hot subspace, its feasibility probability depended only on capacity and forbidden pairs, and it stayed between 0.722 and 0.795 across depths. The X mixer started with almost all of its probability on invalid bitstrings ($P_{\text{feas}} = 0.074$ at $p = 1$) and needed every added layer to recover feasibility. With 32 shots, the XY mixer at $p = 4$ found the optimal plan with probability 88.4%, against 63.7% for the X mixer.

CVaR training raised the X mixer's P_{opt} to 0.060 at $p = 4$, because it rewards the low-energy tail even when most of the state is infeasible. For the XY mixer, CVaR lowered the mean P_{opt} at $p = 4$ to 0.149 but raised success@32 from 88.4% to 98.1%. Because success@N grows sublinearly with P_{opt}, a higher mean success with a lower mean P_{opt} means that CVaR left fewer instances with a very small P_{opt}. If the plan is chosen as the best of a batch of shots, as it would be in practice, CVaR is the better objective for both mixers. Deeper circuits needed more evaluations: 750 on average for the XY mixer at $p = 4$, including the restarts at lower depths.

B. Instance Difficulty

The means in Table I hide a wide spread between instances (Table II). With the mean objective at $p = 4$, the XY mixer's P_{opt} ranged from 0.002 to 0.796. It was 0.284 on the 18 instances where every group could use its cheapest shelter, and fell to 0.111 on the 12 instances where capacity forced at least one group elsewhere. In these instances the optimum is a less obvious plan and has to be separated from cheaper plans that overload a shelter. The X mixer was uniformly poor (0.041 and 0.039), so capacity did not change its result much. CVaR changed the shape of the XY distribution more than its mean: it raised the worst instance from 0.002 to 0.074 and lowered the best from 0.796 to 0.439. This explains why CVaR improved success within 32 shots while lowering the mean P_{opt}. Fig. 3

TABLE I
QAOA ON 30 TWIN INSTANCES (12 QUBITS), MEAN OVER INSTANCES

Mixer	Objective	p	P_{opt}	P_{feas}	succ@32 (%)	Evals
X	mean	1	0.004	0.074	12.2	96
X	mean	2	0.011	0.249	27.6	479
X	mean	3	0.024	0.439	48.2	721
X	mean	4	0.040	0.547	63.7	750
XY	mean	1	0.054	0.734	67.1	272
XY	mean	2	0.093	0.722	77.8	741
XY	mean	3	0.143	0.743	81.9	750
XY	mean	4	0.215	0.795	88.4	750
X	$\text{CVaR}_{0.1}$	1	0.005	0.132	14.6	108
X	$\text{CVaR}_{0.1}$	2	0.019	0.294	44.0	577
X	$\text{CVaR}_{0.1}$	3	0.037	0.356	65.7	728
X	$\text{CVaR}_{0.1}$	4	0.060	0.376	78.4	706
XY	$\text{CVaR}_{0.1}$	1	0.067	0.720	82.7	241
XY	$\text{CVaR}_{0.1}$	2	0.099	0.712	93.9	413
XY	$\text{CVaR}_{0.1}$	3	0.127	0.672	97.3	343
XY	$\text{CVaR}_{0.1}$	4	0.149	0.664	98.1	284

Uniform one-hot sampling: $P_{\text{opt}} = 0.0169$. Evals = objective evaluations at depth p.

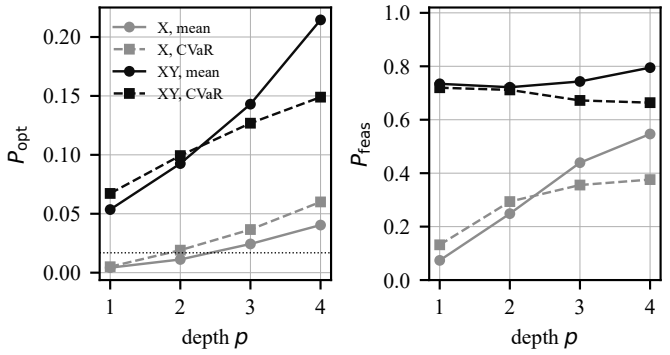


Fig. 2. Probability of the optimal plan (left) and of any feasible plan (right) against depth. The dotted line is uniform sampling of one-hot plans.

shows the individual instances: the XY results with the mean objective include a few instances with P_{opt} near zero, all of them capacity-bound, and CVaR removes that tail.

C. A Worked Instance

Table III shows the first instance (seed 42) after the disruption. Shelter A has only 19 seats left, so groups G3 and G4, with 21 and 20 people, cannot use it at all, and G1 can use it only on its own. Of the 81 one-hot plans, 29 are feasible. The optimal plan, written as the shelter of G1–G4, is ABCB at 80.86 minutes, and the greedy rule also finds it. The runner-up, ABBB, costs 80.88 minutes, only 0.02 minutes more.

Table IV lists the most likely measurements of the trained depth-4 circuits. The XY circuit placed 0.319 of its probability on the optimum and 0.303 on the runner-up, and almost all of the rest on other feasible plans (total 0.982). The X circuit put only 0.049 on the optimum and 0.681 on feasible plans. The near-tie between the two best plans is typical of these instances: travel times of alternative routes are close, so the energy gap that QAOA must resolve is small, and a sampler that finds the runner-up has still found a good plan. This is

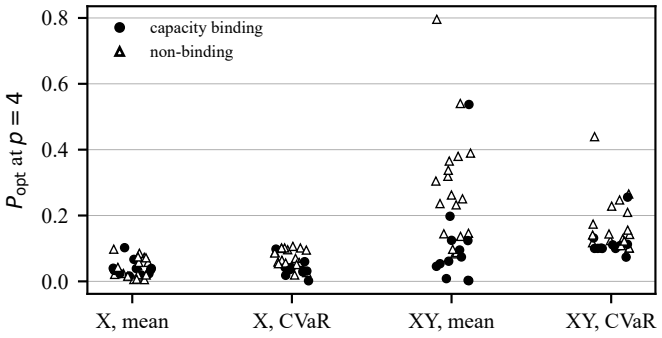


Fig. 3. P_{opt} at depth 4 for each of the 30 instances. Filled circles: capacity changes the optimal plan; open triangles: it does not.

TABLE II
P_{opt} AT $p = 4$ BY INSTANCE TYPE AND ITS SPREAD OVER INSTANCES

	Mean		Over all instances		
Variant	non-binding	binding	min	median	max
X, mean	0.041	0.039	0.005	0.035	0.102
X, CVaR	0.074	0.039	0.002	0.057	0.106
XY, mean	0.284	0.111	0.002	0.145	0.796
XY, CVaR	0.170	0.117	0.074	0.120	0.439

binding = capacity changes the optimal plan (12 instances); non-binding: 18.

also why success within a batch of shots is a more useful measure than P_{opt} alone.

D. Encoding and Circuit Resources

The minimiser of (2) was the true optimum in 93.3% of the main instances. On average an instance had 2.0 overloaded pairs. Per layer, the X-mixer cost unitary needed 14.0 two-qubit ZZ rotations, while the XY variant needed 2.0, because the one-hot penalty disappears. The XY mixer itself adds three two-qubit XY rotations per group and layer, and the W-state preparation adds a fixed overhead at the start [12]. Table V summarises the per-layer gate budget for the 12-qubit instances. The two variants need exactly the same number of two-qubit gates per layer: the 12 one-hot couplings of the X variant become 12 XY rotations. The XY mixer's advantage in Table I therefore does not come from a larger circuit. Compiled to CNOT gates, each ZZ rotation and each XY rotation needs two CNOTs, so one layer needs about 28 CNOTs and a depth-4 circuit about 112, plus the W-state preparation for the XY variant. These counts assume all-to-all connectivity; routing on a device with nearest-neighbour coupling would add SWAP gates.

E. Sensitivity to Gate Noise

All results so far are noiseless. We estimated the effect of gate errors in two ways. The first is a global depolarising model applied to all 30 trained circuits: if every two-qubit gate fails with probability ϵ, the circuit returns its ideal state with probability $F = (1-\epsilon)^{N_2}$ and a uniformly random bitstring otherwise, so

$$P_{\text{opt}}^{\epsilon} = F\,P_{\text{opt}} + (1-F)\,\frac{n_{\text{opt}}}{2^n}, \tag{7}$$

TABLE III
WORKED INSTANCE (SEED 42): ROUTE COST c_{gs} (MIN) AFTER THE DISRUPTION; – MARKS A FORBIDDEN PAIR

Group	n_g	Shelter A	Shelter B	Shelter C
G1	18	16.62	23.58	28.89
G2	13	21.54	20.37	24.97
G3	21	–	20.31	20.30
G4	20	–	23.58	28.89
Seats left		19	76	54

TABLE IV
FIVE MOST LIKELY PLANS OF THE TRAINED XY CIRCUIT FOR THE WORKED INSTANCE, AND THEIR PROBABILITY UNDER THE X CIRCUIT

Plan (G1–G4)	Cost (min)	XY prob.	X prob.	
ABCB	80.86	0.319	0.049	optimal
ABBB	80.88	0.303	0.049	
BBCB	87.82	0.060	0.035	
BBBB	87.83	0.057	0.035	
ACCB	85.47	0.046	<0.035	

where n_{opt} is the number of optimal bitstrings. We count 14.0 two-qubit gates per layer for both mixers (Table V) and charge the XY variant four extra two-qubit gates per W state. With this model the XY mixer kept $P_{\mathrm{opt}} = 0.104$ at $\epsilon = 0.01$, against 0.023 for the X mixer, and depth 4 remained the best choice for both up to $\epsilon = 0.02$; only the CVaR-trained XY circuits did better at depth 2 at that error rate. The ranking does not change for any W-state charge from two to six gates.

The second is a gate-level simulation by quantum trajectories on the first 10 instances, retrained with the same procedure. In each of 200 runs per circuit, every two-qubit gate, including the one-hot and overload couplings, the XY mixer pairs and the W-state preparation, fails with probability ϵ, and a failure applies one of the 15 non-identity two-qubit Pauli operators at random after the layer that contains the gate. Single-qubit errors are ignored. Table VI compares the two models at depth 4 on these instances, whose noiseless P_{opt} was 0.172 for the XY mixer and 0.046 for the X mixer. At $\epsilon = 0.01$, of the order reported for two-qubit gates on current superconducting processors, the XY mixer kept $P_{\mathrm{opt}} = 0.105$, 3.2 times the X mixer's 0.033. The global model predicted only 0.086 for the same circuits. It is pessimistic because many Pauli errors, for example Z errors late in the circuit, leave the measurement distribution almost unchanged, whereas the global model replaces the whole state. The advantage of the constraint-preserving mixer survives both noise models at every error rate we tried. Fig. 4 shows the trajectory results for every depth. Deeper XY circuits stayed ahead of shallower ones at every error rate, because each added layer raised P_{opt} more than its extra gates cost.

F. Classical Baselines and Scaling

Exhaustive search over the 81 one-hot plans took 0.90 ms per instance and is exact. The greedy rule produced a feasible plan in 100.0% of instances and the optimal one in 93.3%. A single short simulated-annealing run (60 sweeps) on the same QUBO reached the optimum in only 10.0% of instances and a feasible plan in 83.3%, in 2.3 ms. It is a weak baseline at this budget, and more restarts would close the gap at negligible cost. Training one QAOA configuration up to $p = 4$ took several seconds of simulation. At this size, QAOA is not competitive on time, and the comparison is about solution quality per circuit, not speed.

TABLE V
GATES PER QAOA LAYER, 12-QUBIT INSTANCES (MEAN OVER INSTANCES)

Gate	X mixer	XY mixer
Cost: single-qubit R_Z	12	12
Cost: two-qubit ZZ	14.0	2.0
Mixer: single-qubit R_X	12	0
Mixer: two-qubit XY	0	12
Initial state	$H^{\otimes 12}$	$\lvert W_3\rangle^{\otimes 4}$

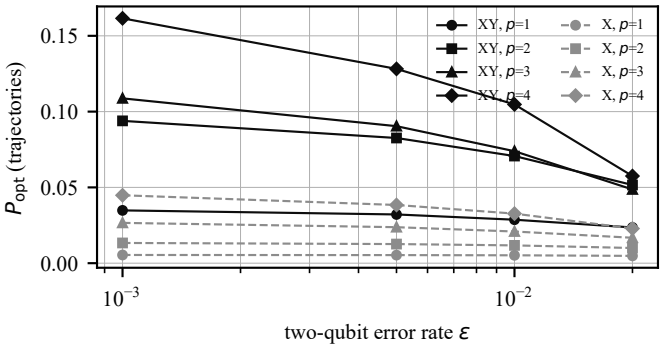


Fig. 4. Gate-level trajectory estimate of P_{opt} against the two-qubit error rate for each depth (10 instances, 200 runs per circuit).

Table VII shows the scaling run. As G grows, the number of one-hot plans grows as 3^G and P_{opt} for uniform sampling falls to 0.0017 at 18 qubits. The XY mixer at $p = 2$ kept $P_{\mathrm{opt}} = 0.006$ at that size, while one training run took 99.4 s of simulation, compared with 8.14 ms for exhaustive search. Statevector cost doubles with each qubit, so exact simulation of this kind stops being practical well before the problem becomes hard classically. Fig. 5 plots the same data on a log scale. The XY mixer stayed above uniform one-hot sampling at every size, from 0.694 at 6 qubits to 0.006 at 18. The X mixer at $p = 2$ fell below uniform one-hot sampling from 12 qubits on (0.0009 at 18 qubits), because most of its probability sat on bitstrings that are not plans at all: its feasibility probability at 18 qubits was 0.051, against 0.271 for the XY mixer. One of the ten 18-qubit instances had no feasible plan and was excluded.

VII. DISCUSSION AND LIMITATIONS

Three points follow. First, building the constraint into the mixer helped more than any change to the training objective, although CVaR still improved the chance of finding the optimum within a fixed number of shots. For assignment problems with one-hot structure, the XY ansatz should be the default,

TABLE VI
P_{opt} AT DEPTH 4 UNDER TWO-QUBIT NOISE: GATE-LEVEL TRAJECTORIES VS. GLOBAL MODEL (10 INSTANCES)

	X mixer		XY mixer	
ϵ	traj.	global	traj.	global
0.001	0.045	0.044	0.162	0.161
0.005	0.038	0.036	0.128	0.121
0.01	0.033	0.028	0.105	0.086
0.02	0.023	0.016	0.058	0.043

traj. = mean over 200 Pauli-error trajectories; global = model (7).

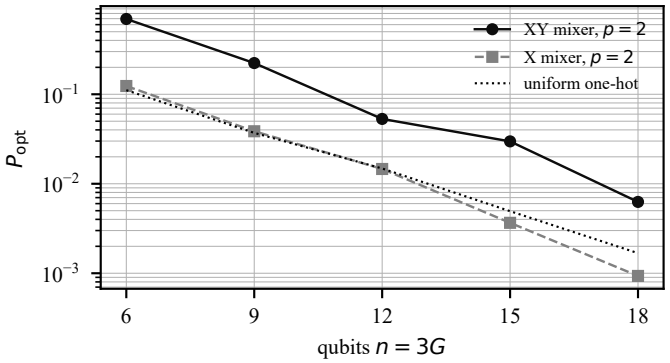


Fig. 5. Probability of the optimal plan against problem size at depth 2 (log scale). The dotted line is uniform sampling of one-hot plans.

and comparisons that use only the transverse-field mixer understate what QAOA can do. Second, the cost shift and penalty normalisation were necessary. Without them the energy is dominated by penalties and the optimiser learns feasibility but not cost. Third, classical methods solve these instances exactly in milliseconds. A realistic city-scale assignment with tens of groups and shelters would need hundreds of qubits and deep circuits, which is beyond the noise levels reported for current processors [15], [17].

The study has clear limits. The main results are noiseless, and the noise study in Section VI-E ignores single-qubit and correlated errors, so the reported probabilities are upper bounds for hardware. The instances are small and come from one synthetic network, and accessibility constraints never bound because all three shelters are accessible. The pairwise capacity encoding is approximate in general. The XY mixer was Trotterised in a single fixed order, and we did not compile circuits to a hardware gate set or count depth after routing. Finally, the angle optimiser had a limited budget, so better angles may exist at higher depths.

For practitioners who want to try such formulations on hardware, our results suggest a short list of choices. Encode one-hot constraints in the mixer, not in penalties. Remove the per-group cost baseline and scale the Hamiltonian so that feasible energies span a unit interval, otherwise the angles must resolve differences that are tiny next to the penalties. Train on CVaR when the plan will be chosen as the best of a batch of shots. Finally, keep a classical exact or heuristic solver in the loop to verify every sample, because even the best variant here placed most of its probability on suboptimal plans for capacity-bound instances.

TABLE VII
SCALING WITH THE NUMBER OF GROUPS ($p = 2$, MEAN OBJECTIVE)

G	n	inst.	unif.	X P_{opt}	XY P_{opt}	XY P_{feas}	XY s	exact ms
2	6	10	0.111	0.124	0.694	0.995	2.4	0.13
3	9	10	0.037	0.039	0.223	0.895	2.0	0.22
4	12	10	0.015	0.015	0.053	0.670	2.1	0.65
5	15	10	0.005	0.004	0.030	0.471	5.7	2.30
6	18	9	0.002	0.001	0.006	0.271	99.4	8.14

unif. = uniform one-hot P_{opt}; XY s = simulated training time; inst. = feasible instances.

Several threats to validity remain. All variants were trained with the same optimiser budget and the same random starts, so the comparison between mixers is fair, but a larger budget could change the ranking at depth 4, where evaluations reached the cap of 750. The instances share one network and one disruption, so the binding and non-binding groups differ only in seeds, not in structure. The success@N metric assumes independent shots from an exact distribution. On hardware, shot noise in the training objective and device noise would both lower it. Finally, the exhaustive reference makes the optimum known, which a real deployment would not have; the verification step we recommend only checks feasibility.

Future work will add single-qubit and correlated noise, transfer angles between instances, compare against slack-variable capacity encodings and test the XY ansatz on hardware through a cloud service.

VIII. CONCLUSION

On shelter-assignment instances drawn from a flood digital twin, a constraint-preserving ring-XY mixer gave QAOA a much higher probability of the optimal plan than the standard transverse-field mixer and kept most of the state on feasible plans at every depth. CVaR training narrowed the gap for the standard mixer but did not close it, and it made success within 32 shots more reliable for both mixers. Exhaustive search solved the same instances exactly in about a millisecond, so the results show which formulation choices matter for future hardware experiments, not a present-day advantage.

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